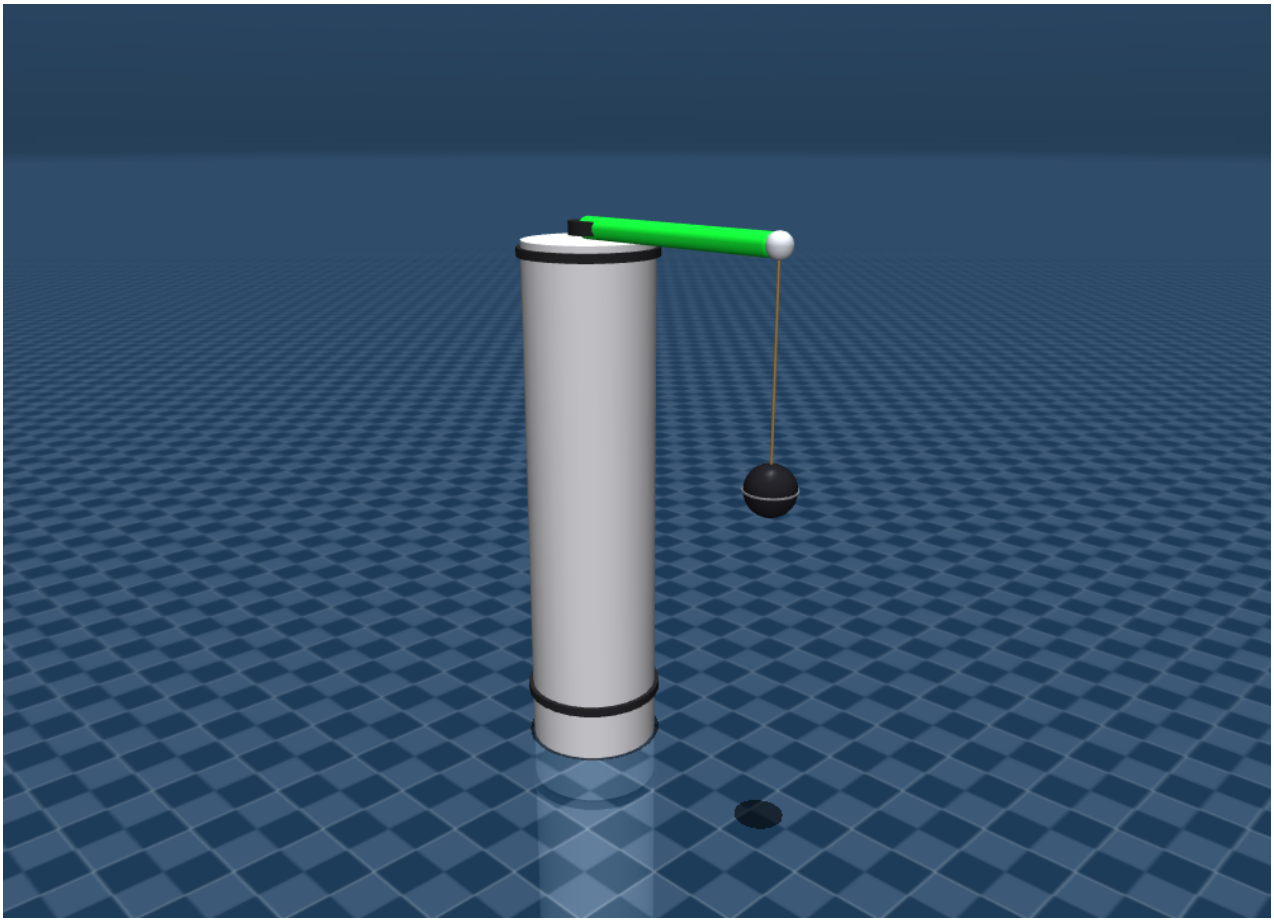
Yaw-actuated pendulum control

PID and Double DQN in MuJoCo

An illustrated analysis of Project_alireza_rezaei_404012012

Alireza Rezaei | 8 October 2026

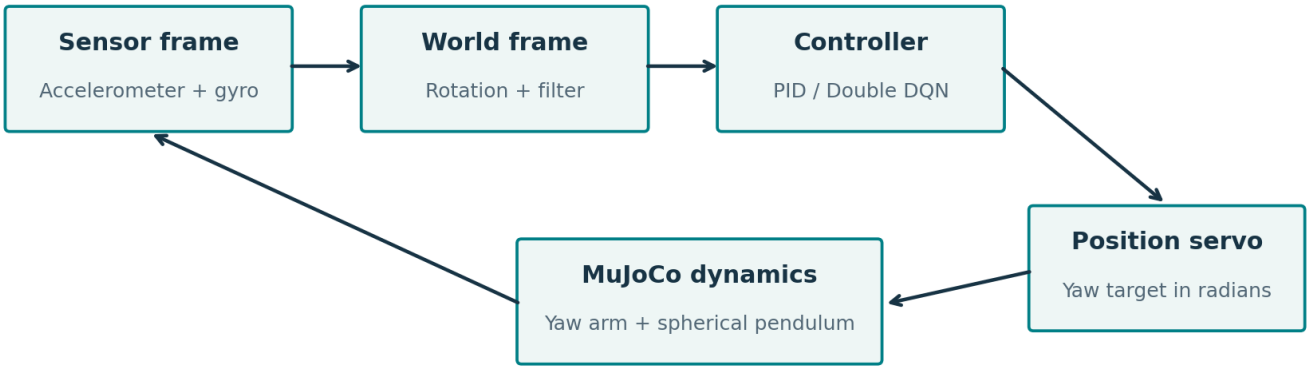


Source review	Fresh simulation	Learning path
Original code, model, 15-page report, plots and weights	8 matched runs; 12 seconds each; zero simulator warnings	7 regressions passed; 30 short training episodes completed

The project connects a physical simulation with classical feedback and value-based learning. This review makes the workflow reproducible and separates historical artifacts from measured evidence. The current gains and inherited policy need further validation before any stabilization claim.

1 / The mechanism and the question

Can a yaw command reduce motion of a ball hanging from a rotating arm after a force pulse? One actuator influences an unactuated spherical pendulum, making this a nonlinear, underactuated control problem.



Schematic control loop; one yaw actuator, no direct torque action

Model element	Specified value / interpretation
Kinematic chain	world -> stand -> green_link -> rope_rigid
Arm / attachment	1.00 m capsule, 2.00 kg; pendulum attachment at 1.05 m
Suspended segment / ball	0.96 m rigid segment; 0.50 kg ball
Degrees of freedom	1 yaw hinge + 3 rotations at the ball joint; 1 actuator
Gravity / integration	(0, 0, -9.81) m/s^2; implicitfast; 0.0005 s physics step
Position servo	Gain 350; damping ratio 1; force range +/-300
Yaw command	Target position, not direct torque; limits +/-170 degrees

The word rope refers to a single rigid capsule. It cannot bend, stretch or become slack. Passive joint damping is present, so holding the yaw target at zero is an essential baseline. The simulation does not establish real cable or hardware behavior.

The diagram is a schematic. The cover image is an offscreen render of the actual maintained MJCF model. Model specifications come directly from the supplied XML.

2 / Measurements and PID feedback

From local sensors to a bounded command

The site accelerometer and gyro are transformed to world coordinates. The maintained filter updates once per physics step; observation reads are pure. Undisturbed warm-up lasts 0.25 s and establishes the horizontal baseline before the kick begins.

$$\mathbf{a}_W = R_{W \leftarrow S}\mathbf{a}_S, \quad \bar{\mathbf{a}}_k = (1-\alpha)\bar{\mathbf{a}}_{k-1} + \alpha \mathbf{a}_{W,k}$$

$$\psi_{k+1}^{cmd} = \mathrm{clip}\left(\psi_k^{cmd} + \mathrm{clip}(\psi_k^{target} - \psi_k^{cmd},\ -\dot{\psi}_{max}\Delta t_c,\ \dot{\psi}_{max}\Delta t_c),\ \psi_{min}, \psi_{max}\right)$$

$$a^* = \arg\max_a Q_\theta(s', a), \quad y = r + \gamma(1 - d_{terminal})Q_{\theta^-}(s', a^*)$$

Equations: frame rotation and filtering; bounded/rate-limited yaw target; Double DQN bootstrap. Filter coefficient alpha = 0.15. The common control interval is 0.005 s.

The inherited acceleration-feedback law

PID regulates world-X acceleration with gains Kp = 0.35, Ki = 0.04 and Kd = 0.08. The output maps to a yaw target through a negative sign and radians-to-degrees conversion. A 0.5 m/s² deadband, integral clamp of 30 and saturation-aware integration constrain the command.

Implementation point	Why it matters
Derivative timing	Use the 5 ms controller period. The archived code divides by the 0.5 ms physics step.
Calibration	A rest baseline must precede the force pulse; the historical calibration overlapped it.
Control authority	One yaw input cannot independently regulate both horizontal axes. The influence depends on configuration.
Interpretation	Correct timing and units do not make the inherited gains a validated stabilizing controller.

The vertical sensor channel retains the gravity-related reading. Its sign should be established from an actual rest test instead of assumed from the original report. The benchmark uses horizontal acceleration and separately logs pendulum tilt.

3 / What makes this Double DQN

The online network selects the next action and the target network evaluates that action. This is Double DQN, despite the historical filename using DQN. The network is an 8-256-256-2 MLP with ReLU hidden layers.

Component	Implementation
Observation	World acceleration (3), world gyro (3), normalized actual yaw and normalized target-command rate
Actions	Left/right requests of 1.5 degrees; rate cap realizes at most 1.2 degrees per 5 ms step
Original budget	1,500 episodes; maximum 800 decisions per episode
Discount / optimizer	Gamma 0.99; Adam learning rate 0.0003
Replay / batches	Capacity 200,000; batch size 256; learning begins at 2,000 transitions
Target / exploration	Synchronize every 1,000 transitions; epsilon 1.00 to 0.05 over 120,000 transitions
Loss / clipping	Smooth L1 loss; gradient norm capped at 10

Reward and episode ending

Reward penalizes horizontal acceleration via -tanh(a_xy / 10), yaw magnitude via -0.05 |yaw| / 90 and target changes via -0.02 |delta yaw| / 1.5. After the pulse, acceleration below 3 m/s^2 earns +1. Forty consecutive qualifying decisions earn +200 and success termination.

The modeling limits

A 0.20 s acceleration window is a proxy, not a stability proof. Target-command rate is not measured joint velocity; the gyro supplies rotational measurements. The observation has not been proven to be a sufficient Markov state, and the discrete action set has no explicit hold action.

A true terminal transition masks bootstrapping. An 800-step time limit is truncation and retains it. Changing units, filtering and normalization changes the inherited policy inputs; a fresh trained policy is needed for comparison in the maintained environment.

4 / Audit findings and maintenance

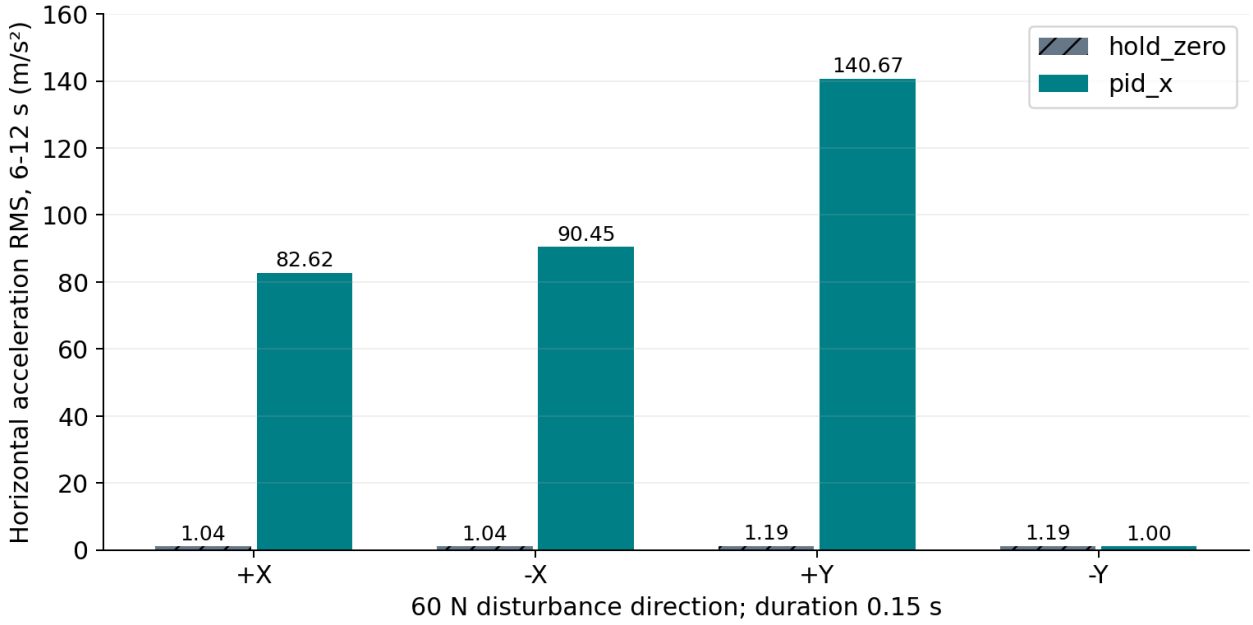
The original source and report remain unmodified in the archive. The maintained code addresses concrete reproduction issues; the new protocol is not presented as identical to the historical experiment.

Supplied artifact finding	Effect	Maintained change
Target range -170 to 170	Compiles as radians, unlike the degree-specified hinge	Explicit +/-2.967059728 rad
PID divides derivative by physics dt	About 10x derivative scale at a 200 Hz update	Use actual 0.005 s interval
Calibration overlaps kick	Disturbed signal becomes the rest baseline	Calibrate before disturbance
RL filters endpoint twice	Observation and reward use different filter state	Filter once per physics step
One done flag for success/time limit	Time-limit target incorrectly loses bootstrap	Separate terminal and truncated
Success checked during kick	Can reward early timing rather than post-kick settling	Count after kick completion
Four-actuator UI vs one-actuator XML	Historical UI is incompatible with model	One-yaw viewer command
Large legacy constraint allocation	Initial benchmark hit a memory allocation failure	Explicit 64 MB arena
No checkpoint/log writer in training	Supplied weights lack training provenance	Save metadata, weights and CSVs
Working-directory / launch assumptions	Scripts may open viewers during import	Portable model path; explicit CLI

The original PID rate cap was 180 degrees/s and RL used 240 degrees/s. The new benchmark uses a common 240-degree/s cap. The original checkpoint diagnostic retains historical control logic and only changes memory allocation.

5 / The fresh matched benchmark

Eight deterministic runs: two conditions for each of +X, -X, +Y and -Y. After 0.25 s undisturbed warm-up, apply 60 N for 0.15 s and observe 12 s after force onset. Both conditions share the same model, filter/control cadence, limits and rate cap. PID uses inherited gains with corrected timing.



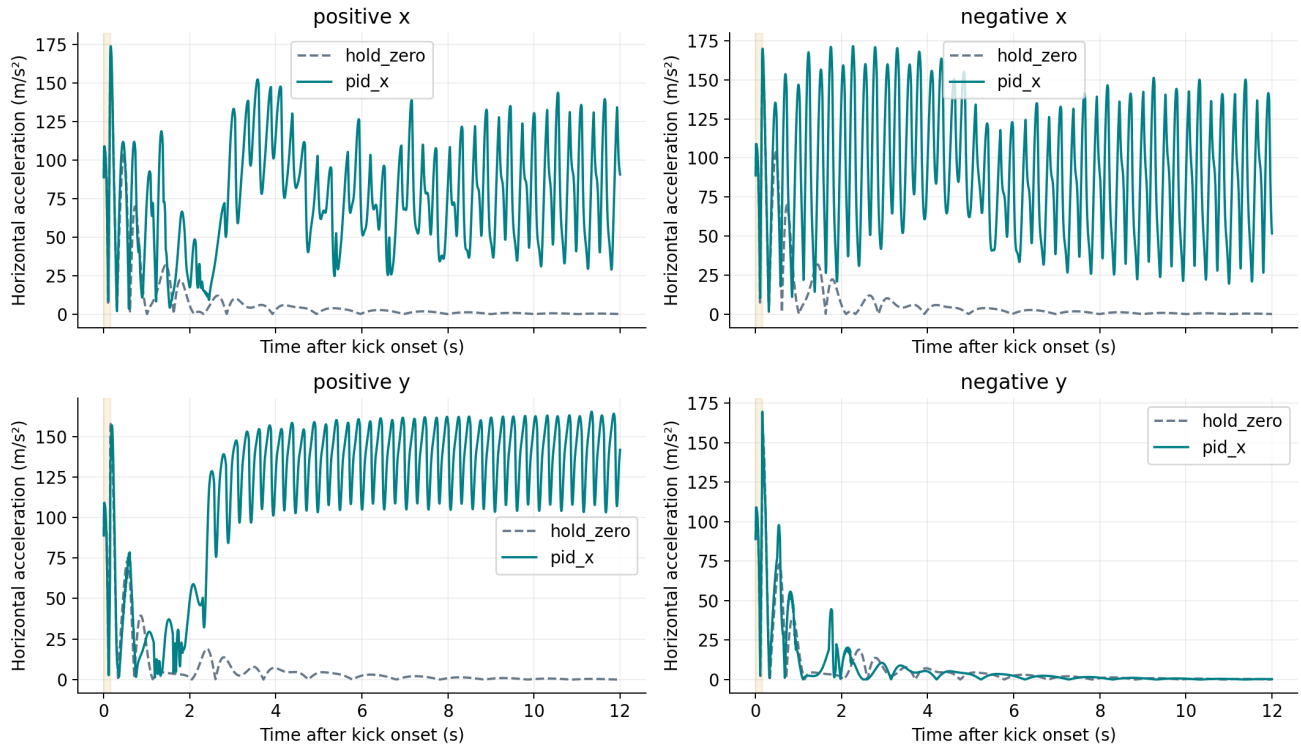
Measured from saved trajectories; RMS window = 6-12 s. These are fixed scenarios with no repeated-trial uncertainty estimate. Lower acceleration does not by itself imply stable equilibrium.

Kick	Hold RMS	PID RMS	Final tilt: hold / PID
+X	1.04	82.62	0.59 / 54.33
-X	1.04	90.45	0.59 / 131.20
+Y	1.19	140.67	0.15 / 63.55
-Y	1.19	1.00	0.15 / 1.76

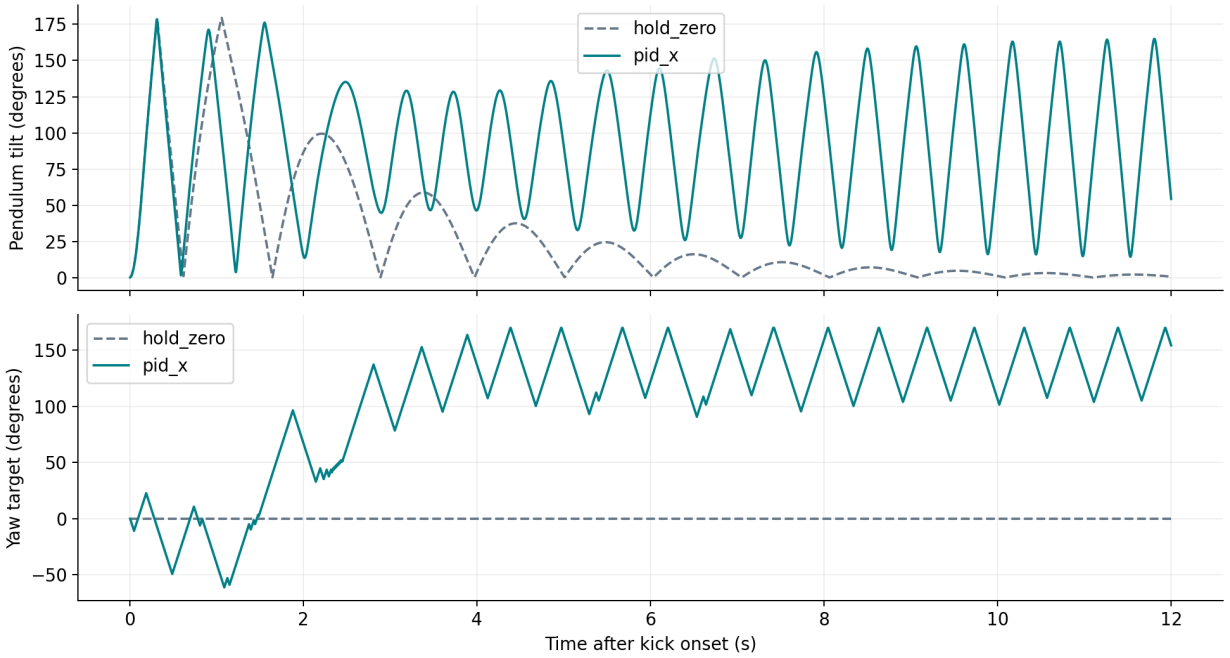
The inherited PID gives higher late acceleration in three of four directions. In -Y its RMS is slightly lower, but its final tilt is larger. All runs include nearly full pendulum rotations, so this is outside a small-angle regime. No trained DQN is ranked against corrected PID here.

Acceleration: m/s^2. Tilt: degrees from downward vertical. All eight runs reported zero simulator warnings. Numerical execution is distinct from physical validity and successful control.

6 / Read the trajectories, not one score



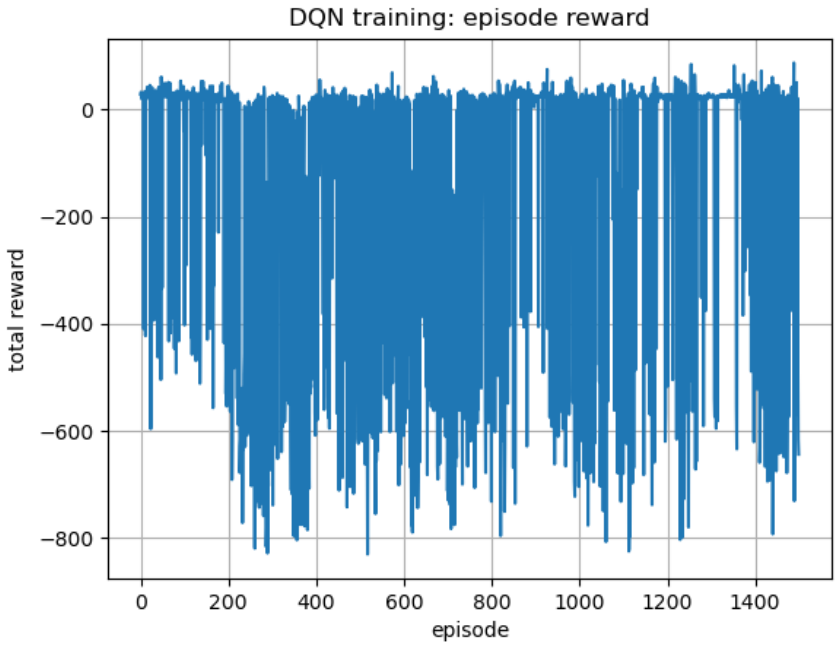
Horizontal acceleration for all four force directions. Shading marks the 0.15 s pulse. Dashed gray holds yaw at zero; teal uses inherited PID gains with corrected timing. Full-horizon logging continues after any short success-proxy window.



+X case: pendulum tilt and yaw target. The PID reaches the target limit and sustains motion. The first threshold window is not called a permanent settling time. Raw trajectories and exact settings are saved with the repository.

7 / Historical evidence and checkpoint

Seven supplied plots show training and evaluation traces. Their raw numerical data and exact link to the checkpoint are missing. They remain historical visual evidence; this analysis does not digitize them into new performance claims.



Historical episode-reward plot from the supplied archive. It is not a fresh training result from the maintained code.

Checkpoint diagnostic	Observed result
Network tensors	Compatible with 8-256-256-2; loaded with weights_only=True
Replay horizon	800 decisions; 4.025 s simulated time
Total reward / final acceleration	-784.07 / 48.70 m/s²
Success / numerical warnings	Stability counter 0; warning count 0
Interpretation	Ended at the archived time limit; success was not achieved in this diagnostic

The replay uses original Python logic. Only the model arena is reduced to 64 MB. It does not reproduce the original training or provide a fair comparison with the corrected PID. Checkpoint training configuration and optimizer history are absent.

The original RL viewer appends across resets while simulation time restarts; a combined evaluation trace can therefore contain time discontinuities.

8 / Reproduction and verification

Verified locally with Python 3.12.9, MuJoCo 3.5.0, NumPy 1.26.4, Matplotlib 3.10.0 and PyTorch 2.4.0 on Windows. Install from a checkout; the maintained package uses the adjacent models directory.

```
python -m pip install -e .
python -m unittest discover -s tests -v
pendulum-control benchmark --out results/local/benchmark
pendulum-control train --episodes 30 --max-steps 200
python scripts/inspect_original.py
```

Verification	Coverage / evidence
7 passing regression tests	Compiled radians and rate limits; pure observations and repeatable reset; episode endings; PID timing and anti-windup; Double DQN targets; replay wrap
8 full benchmark runs	Raw CSVs, summary metrics, plots, model hash, actual configuration and version manifest
30 short training episodes	6,000 transitions; 21 episode records with optimizer loss; weights/metadata written locally
Original archive integrity	15 original files preserved with SHA-256 hashes
Visualization	Actual MJCF offscreen scene; measured plots; editable SVG diagrams

What these checks establish

The simulator, controller interfaces, replay sampling, gradient updates, target synchronization and checkpoint-writing paths execute. The short training check does not establish convergence or policy quality. Its weights are not promoted as a validated controller.

Evidence boundaries

The original 1,500-episode training was not rerun. There is no independent learned-policy evaluation or hardware trial. Future runs should use local output folders so the checked-in evidence remains traceable. Core version pins are supplied, rather than a complete transitive environment lock.

9 / A credible next experiment

Step	Next deliverable
1. Define the target	Angle, angular velocity, yaw motion and persistent settling criteria, beyond acceleration alone
2. Establish baselines	Hold-zero and interpretable feedback controllers under matched conditions; analyze local yaw control authority
3. Separate development and testing	Tune on one disturbance/initial-state set; reserve forces and directions for evaluation
4. Retrain the policy	Corrected environment, full metadata, multiple seeds; investigate a hold action and angle/velocity observations
5. Report generalization	Fixed horizons, repeated trials, uncertainty, constraint violations and matched compute budgets
6. Review physical fidelity	Contact, damping, sensor noise and flexible cable assumptions before any hardware transfer

The strongest present outcome is a transparent, executable research prototype with an auditable explanation. The measured failures are useful design evidence. A broad claim that PID or RL solves stabilization would require the experiments above.

References and source evidence

MuJoCo XML reference: sensors, actuator limits and arena allocation

MuJoCo modeling: kinematic frames and compilation

Van Hasselt, Guez and Silver: Double Q-learning

Project-specific specifications, findings and measurements are supported by the original source/report, archived checkpoint diagnostic and fresh checked-in CSVs. The detailed Markdown analysis includes file-level evidence and reproduction instructions.

Original project: Alireza Rezaei. Original academic artifacts remain intact. No additional software reuse license is granted by this report.